FACULTY OF ENGINEERING TECHNOLOGY

Subject: Comm.& Data Transmission — Final Exam summer 24/25

Name:..... — Duration: 90 min

QI: choose the correct answers:

(Please write your answers in the table only; otherwise, your answers will be neglected) (34 marks)

Question	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*	*12*	*13*	*14*
Answer														
Question	*15*	*16*	*17*											
Answer														

1. The twisted pair cable in which metal casing improves the penetration of noise or crosstalk is called
   A. insulated twisted pair cable
   B. Shielded twisted pair cable
   C. Unshielded twisted pair cable
   D. Both A and B
2. Multiplexing combines signals from several sources to achieve
   A. Data rate management
   B. interleaving
   C. TDM efficiency
   D. bandwidth efficiency

3. The synchronous Time-Division Multiplexing (TDM) is not efficient because of
   A. higher data rate
   B. infinite frames
   C. empty slots
   D. Both A and C

4. The multiplexing technique that is used when the data rate of an input line is a multiple of the others is called
   A. multilevel multiplexing
   B. multiple-slot allocation
   C. pulse stuffing
   D. Asynchronous TDM

5. TDM is a digital multiplexing technique for combining several low-rate channels into
   A. a group of low-rate channels
   B. one high-rate one channel
   C. two high-rate channels
   D. None

6. Each TV channel has its own bandwidth of
   A. 6 Hz
   B. 6 KHz
   C. 6 MHz
   D. 60 MHz

7. In synchronous TDM, the data rate of the link is n times faster and the unit duration is:
   A. n times shorter
   B. n+1 times shorter
   C. n-1 times shorter
   D. remains the same

8. The frequency difference between the Wavelength-Division Multiplexing (WDM) and Frequency-Division Multiplexing (FDM) is
   A. very high
   B. very low
   C. zero
   D. infinity

9. In OOOVBOVB, V stands for violation, and B stands for
   A. Bit
   B. Baud
   C. Bandwidth
   D. Bipolar

10. The last step in Pulse Code Modulation (PCM) is
   A. Quantization
   B. Sampling
   C. Encoding
   D. Modulation

11. Data encoding is not possible if
   A. $2^m < L^n$
   B. $2^m > L^n$
   C. $2^m = L^n$
   D. $2^m \# L^n$

12. In line-of-sight propagation, very high-frequency signals are transmitted in straight lines directly from the antenna to
   A. sky
   B. earth
   C. antenna
   D. planet

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13. In real life, most digital signals have bandwidth
A. Infinite
B. Zero
C. Constant
D. None of the above

14. Line coding converts a sequence of
A. Frames to digital signal
B. bits to a digital signal
C. Bytes to digital signal
D. Packets of data to digital signals

15. The bandwidth is proportional to the
A. baud rate
B. data rate
C. delay rate
D. bit rate

16. Radio waves are
A. unidirectional
B. omnidirectional
C. bidirectional
D. directional

17. Optical fibers use reflection to guide light through a
A. channel
B. metal wire
C. light
D. plastic



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